

RYAN GREY

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AWS-certified data and operations professional with 10 years in healthcare information systems at a 13-hospital, \$2B health system — built the reporting and dashboards behind \$1B+ in annual billing. Completing the AWS AI certification track with hands-on Amazon Bedrock work.

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services, August 2026 · [verify](#)

AWS Certified AI Practitioner (AIF-C01) — in progress, 2026

Epic Certification — Mountain States Health Alliance



TECHNICAL SKILLS & PROJECTS

Cloud (AWS): EC2, S3, IAM & STS, Cognito, VPC, Lambda, API Gateway, DynamoDB, CloudFront, Route 53, SNS, SES, CloudWatch, EventBridge, Cost Explorer & Budgets, SSM Parameter Store, KMS, Amazon Bedrock, SageMaker (fundamentals), AWS pricing & Well-Architected Framework

Build & Languages: Swift / SwiftUI, Python, JavaScript, Bash; Git & GitHub Actions CI/CD, AWS SigV4 request signing, Ed25519 signature verification, cost engineering (FinOps)

Data & Reporting: Dashboard design, KPI reporting, database analysis, query writing, SharePoint administration, Google Apps Script automation

Health IT: Epic, Cerner, SMS Envision, 3M Coding, Chargemaster, PayNav, Audit Logix; HIPAA / PHI compliance

- **ryangrey.dev** (AWS, live 2026 · ryangrey.dev · [source](#)) — deployed end-to-end on AWS: private S3 origin behind CloudFront with Origin Access Control, ACM TLS, Route 53 hosted zone and alias records; migrated registration from GoDaddy with DNSSEC DS-record teardown sequenced before the nameserver cutover
- **ryangrey.dev delivery & monitoring pipeline** (GitHub Actions · IAM OIDC · CloudFront Functions · CloudWatch · SNS · Lambda · SES · EventBridge, 2026) — push-to-main deploys assume a least-privilege IAM role via OIDC federation with zero stored AWS keys, then verify the live page byte-matches the commit; diagnosed GitHub's undocumented immutable-ID subject claims by decoding the runner's JWT; security headers injected at the edge by CloudFront Functions with a path-branched CSP — `script-src 'none'` on the main page; a CloudFront 5xx alarm delivers through a Lambda-to-SES pipeline signing DKIM as ryangrey.dev (DMARC passes on DKIM alignment where SPF cannot), and an EventBridge Scheduler heartbeat force-fires that alarm monthly through a path sharing no failure modes with the pipeline it tests
- **Cloud Watchtower** (Swift/SwiftUI · CloudWatch · Cost Explorer · IAM/STS, 2026) — native macOS menu-bar monitor for site health and spend, built with **zero third-party dependencies** — including a hand-rolled AWS SigV4 signer across two request protocols; authenticates by assuming a scoped read-only role on short-lived STS credentials rather than holding a deploy key; cache-first polling design costs ~\$0.12/month against roughly \$8.50 for a conventional poll-everything build (60×+ cheaper) and caught 4 of 4 alarm state transitions on first test; in service it surfaced a sustained 4xx error rate that the existing 5xx-only alarm was structurally unable to see
- **Retrieval-augmented (RAG) chatbot** (Amazon Bedrock · Lambda · API Gateway · DynamoDB, live 2026 · ryangrey.dev/ask) — production chatbot grounded in a document corpus: Amazon Nova Lite via cross-region inference profile, Titan v2 embeddings with in-process retrieval — a deliberate no-vector-database design documented as linear in corpus size; treats cost as a security property — cheapest-first checks and fail-closed rate limiting on DynamoDB TTL counters, so an outage denies requests rather than uncapping spend; IAM as the real boundary — no credentials in the browser, Lambda role scoped to 7 ARNs with zero wildcards, prompt injection treated as mitigation, not boundary
- **AIF Study Engine** (Cognito · API Gateway · Lambda · DynamoDB · Amazon Bedrock, live 2026 · ryangrey.dev/quiz · [source](#)) — private spaced-repetition exam-prep tool for the AWS AI Practitioner track: Cognito Hosted UI (PKCE) and an API Gateway JWT authorizer gate every request; Bedrock generates quizzes hard-scoped to completed lessons only — the scope filter is enforced and logged server-side so a leak is visible, not silent; misses persist to DynamoDB on a spaced-repetition ladder; ~\$0.06/month
- **GreyScale** (SwiftUI · Swift Charts · Cognito · API Gateway · Lambda · SQS · DynamoDB · Amazon Bedrock, 2026 · TestFlight) — private iOS app and serverless backend for a four-person family's weigh-ins and calorie tracking, designed, built and shipped solo; five ways to log a meal — typed entry, barcode scan (VisionKit), USDA/OpenFoodFacts search, a plate photo read by Bedrock Nova Lite over an async S3 → SQS → Lambda pipeline, and on-device nutrition-label OCR; 55 days of history imported so the weekly view shipped complete; daily maintenance calories back-solved from intake against the scale over the longest unbroken window; single-table DynamoDB reached only by *Query* on the partition key — no *Scan* in the code or the IAM policy; OAuth 2.0 Authorization Code with PKCE; auth fails closed at both the gateway's JWT authorizer and the handler, which re-checks *token_use*; no endpoint accepts a member id, so every row key derives from the verified token claim; verified against the replaced spreadsheet's 14 weeks rather than against itself
- **greyBot** (Lambda · EventBridge · DynamoDB · API Gateway · SSM/KMS, live 2026 · [source](#)) — serverless Discord bot announcing a raid guild's boss progression and post-raid recap; exactly-once delivery enforced by a DynamoDB conditional write rather than read-then-write, so overlapping invocations cannot double-post to a live channel; two raid teams log to one source with no API field separating them, so team identity is *derived* from who recurs across each tier's first kills, and the classifier posts nothing rather than guess when its signals disagree — the card names individuals; three API fields return an untyped JSON scalar, so parsers were built from captured production responses, surfacing six defects the spec would have hidden, including a table that silently truncates at 200 rows; a */progress* slash command serves over Ed25519-verified HTTP interactions; 341-check offline suite gates every deploy; ~\$0.02/month

PROFESSIONAL EXPERIENCE

Owner — E-Commerce Operations — *Self-employed · eBay & Facebook Marketplace*

2019 – Present

- Operate a resale business across eBay and Facebook Marketplace — research market comps, set pricing strategy, write listing copy, and negotiate sales while managing 40+ concurrent listings
- Completed computer science coursework (Ohio State) and Sabio full-stack bootcamp training concurrently with the business

Revenue Cycle Operations Coordinator — *Ballad Health (formerly Mountain States Health Alliance), Johnson City, TN*

Nov 2014 – May 2019

- Streamlined data collection and reporting for \$1B+ in annual billing across 13 hospitals by authoring standardized templates and centralized dashboards
- Served as sole support for 6 revenue-cycle systems (PayNav, Audit Logix, SharePoint, Chargemaster, US Bank, SCI); built the front-end revenue-cycle dashboard reported daily to CFOs
- Reduced staffing inefficiencies in the billing division by building utilization reporting that senior leadership adopted into revised staffing models
- Managed 3 direct reports and provided oversight for 18 coders; improved KPI performance across Epic, Cerner, SMS Envision, and 3M platforms while maintaining HIPAA/PHI compliance

Revenue Cycle Precertification Coordinator — *Ballad Health (formerly Mountain States Health Alliance), Johnson City, TN*

Nov 2013 – Nov 2014

- Secured prior authorizations across Medicare, Medicaid, and commercial payers; verified medical necessity and coordinated with clinics and utilization review to prevent denials
- Served as the team's sole information-systems support and lead for PayNav, Audit Logix, and the back end of the revenue cycle's chargemaster
- Designed and built the department's precertification tracking database in Microsoft Access — front and back end, used by 40+ reps, supervisors, and managers — and ran it in production with monthly backups and maintenance; its reports went directly to executive leadership

Patient Access Representative — *Ballad Health (formerly Mountain States Health Alliance), Johnson City, TN*

Aug 2009 – Nov 2013

- Processed registration and insurance eligibility (Medicare, Medicaid, BCBS) for up to 100 patients daily; trained new team members across multiple facilities during software launches

EDUCATION & PROFESSIONAL DEVELOPMENT

AWS Skill Builder — AI Practitioner Learning Plan & Exam Prep Plan (~30 hrs), 2026 · Cloud Practitioner exam track, 2026

The Ohio State University (2 semesters, remote) · **East Tennessee State University**, Johnson City, TN — computer science coursework toward B.S.

Sabio Coding Bootcamp — Full-Stack Developer program (3 quarters, remote)